

Revision round #1

Decision for round #1 : **Revision needed**

Revision invited

Review summary by Chris Chambers

Comment 1: I have now received three reviews of your Stage 2 submission. As you will see, the reviews are detailed and constructive (for which I offer my sincere thanks to the reviewers) and there is a positive consensus on the quality of your study and the effort involved in completing such an ambitious field experiment. On my own non-specialist reading, I must also applaud you for the degree of transparency in the reporting and the great effort made to adhere to the principles of Stage 2 RRs. It is clear that performing RRs in this area is incredibly challenging due to the unpredictability of field work, and you have navigated these challenges with skill and integrity.

Response 1: Thank you so much for the kind words and acknowledgment of the effort that went into this research!

Comment 2: The main issue to address in revision is the clarity of mapping between the accepted Stage 1 protocol and the completed Stage 2 report. As the reviewers note, the Stage 2 manuscript necessarily reports a subset of the originally planned jay component: disturbance-resilient California scrub-jays or blue jays could not be sampled so J.Q2 could not be tested as planned, and some achieved samples were smaller than planned. These deviations are already disclosed, but the manuscript would benefit from more explicit signposting to make it immediately clear which research questions and analyses were confirmatory and tested as planned, which confirmatory components could not be tested, and which analyses are exploratory/post hoc. In reading your submission, it occurred to me that a concise “planned-versus-achieved” table could be helpful, listing each Stage 1 question, the planned species/sample/analysis, the achieved sample, whether confirmatory inference was possible, and any exploratory analysis reported instead or in addition.

Response 2: I added text to the Abstract (lines 36-39) and the Introduction (lines 118) indicating that we were not able to test a disturbance resilient jay species. I created the table that you suggested, inserting into the Stage 2 manuscript at lines 438-440. I also moved the description of the unregistered analysis and results to a separate subsection of the Methods (lines 422-437) and Results (lines 485-491), with a clear heading.

Comment 3: At the same time, I want to be clear that I'm not asking you to rewrite the manuscript as a paper focusing only on Florida scrub jays or to remove the Stage 1 framework. Because this is a Stage 2 RR, preserving the structure and logic of the accepted Stage 1 protocol is paramount for transparency and auditability. For this reason, some reviewer comments, particularly those suggesting substantial reframing away from the approved Stage 1 structure, should be interpreted as requests for clearer Stage 2 signposting rather than as requirements to abandon that structure. For example, contrary to the suggestion of one reviewer, J.Q2 should not be deleted – rather, retaining it is in my view the better approach provided it is explicitly marked as a planned question that could not be tested.

Response 3: We appreciate you providing guidance on those types of comments because it can be hard to know how to satisfy peer reviewers while maintaining transparency in the research process.

Comment 4: Please also clarify the achieved sample sizes and the contribution of individuals to each analysis. The reviewers raise reasonable questions about how many birds contributed to baseline reversal learning, serial reversal manipulation, microhabitat observations, food-use observations, and each model. Please state clearly where the achieved sample was below the planned minimum, explain why, and indicate how this affects interpretation. In particular, clarify the analysis involving birds from natural versus human-modified areas, including the implications of the small and unbalanced sample for any inference involving the human-modified group.

Response 4: We understand how this would have been helpful. We created the table, as per your comment above, that has these sample size details included. I also included the achieved sample sizes in the appropriate sections of the Results.

Comment 5: The exploratory comparison between Florida scrub-jays in suburban and natural habitats also needs clearer framing. This analysis is permissible and potentially useful, but it should be consistently labelled as unregistered/exploratory and interpreted cautiously. Please provide a stronger rationale for the analysis and acknowledge relevant interpretive limitations, including differences in microhabitat availability across environments. If you're able to add descriptive information on available microhabitats or proportional use without overcomplicating the paper, that may be useful, but this isn't a requirement.

Response 5: We now distinguish the unregistered analysis more clearly in the Methods, Results and new Table 2. In the Methods section (lines 422-437), I added text to explain the rationale for conducting this analysis. I added to the Results section (lines 492) a figure (Figure 5) containing descriptive information on the proportion of microhabitats and food types used by jays during observations in the suburban and natural sites. In the

discussion in lines 513-514 and lines 531-539 we emphasize caution in interpreting these post-hoc findings.

Comment 6: Several reviewer comments ask for clarification of methodological details, and these should be addressed. As noted in the author guidelines, please also update the Abstract so that it reflects the completed Stage 2 study, including methods, key results, and conclusions, and revise remaining future-tense language inherited from the Stage 1 manuscript where it now refers to completed work.

Response 6: Apologies for the oversight on the Abstract details. Specific methods, key results and conclusions are now added and revised to past tense.

Comment 7: Finally, please address the reviewers' minor comments and copy-edits where appropriate (or rebut accordingly). One point I want to emphasise is that you don't need to revise the approved design or analyses simply because a reviewer would now prefer a different design, outcome metric, or analytic approach. In particular, the approved use of serial reversal learning, the reversal-passing criterion, and the confirmatory outcome measures of number of microhabitats used and number of food items taken will not be relitigated at Stage 2. You may, of course, add clarifications, limitations, or clearly labelled exploratory/sensitivity analyses if you think these improve the manuscript, but the core principle at Stage 2 is to report the approved analyses transparently and interpret them accordingly.

Response 7: Thank you for the time you took to review our Stage 2! We addressed the reviewers' minor comments as you suggested and explained the rationale for the Stage 1 approved design/analyses when appropriate.

Review by Gloriana Chaverri, 21 May 2026 19:48

Comment 8: I have reviewed the post-study stage 2 manuscript "Flexibility training does not increase behavioral diversity of Florida scrub-jays", based on the accepted Peer Community in Registered Reports titled "The role of behavioural flexibility in promoting resilience to human environmental impacts". I found this Stage 2 manuscript to be a strong and appropriate completion of the approved Stage 1 registered report. The study clearly follows the central aims of the registered protocol: testing whether behavioral flexibility in jays can be increased through serial reversal learning, and whether such training translates into broader behavioral diversity in food use or microhabitat use. The manuscript presents the research questions, predictions, methods, and analyses clearly, and the authors distinguish appropriately between confirmatory

analyses and the unregistered analysis conducted when one original question could not be addressed.

In my view, the criteria for approving a Stage 2 Registered Report are met. The authors provide a transparent account of the deviations from the Stage 1 plan, especially the inability to sample disturbance-resilient jay species and the smaller sample available for the full manipulation. These deviations are explained clearly and do not appear to undermine the integrity of the analyses that were possible. The manuscript also reports the results in an outcome-neutral way: the authors do not overstate the negative findings, but instead conclude that flexibility training improved performance on the reversal-learning task without producing detectable increases in food or microhabitat diversity.

Overall, I think the study did what it set out to do within the practical constraints of the system. The conclusions are well aligned with the data, the limitations are acknowledged, and the interpretation remains appropriately cautious. I therefore support approval of this Stage 2 manuscript.

Response 8: Thank you so much for taking time to review our manuscript! We are happy that you found the Stage 2 to be appropriate and clear!

Review by Vedrana Šlipogor, 16 May 2026 13:19

Comment 9: Flexibility training does not increase behavioral diversity of Florida scrub-jays is an interesting post-study manuscript of the programmatic registered report exploring whether flexibility can be increased experimentally by increasing heterogeneity in the environment and whether this increase can help species succeed in human-modified environments. Due to some constraints, the Authors could not test their original idea of flexibility interventions in California scrub-jays or blue jays, resilient species successful in human modified environments, to see how flexibility relates to success and implemented these interventions in Florida scrub-jays, a sensitive species, to determine whether flexibility can be trained, and if such training improves success in human-modified environments. Instead, they compared two Florida scrub-jay populations, one living in suburban areas and one living in natural habitat without human disturbance, which was an unregistered analysis. Data collection thus to an extent followed preregistration plan, and the made changes were clearly indicated in the text (see L48-53). Data analyses also to an extent followed the preregistration plan and were done using adequate statistical techniques.

The manuscript is, however, written with original research ideas and framing, which is difficult to grasp for the naïve reader, and thus I would suggest rewriting it to follow the narrative of the ideas that could be tested in this manuscript, i.e. comparing two Florida scrub jay populations. It is also unclear a bit how many birds participated in which tests/observations/analyses; and why the Authors conducted comparisons on reversal learning between birds in the natural areas and birds in the human modified areas, as the latter included only 1 bird. I also had some trouble following what Authors mean by “behavioural flexibility” as a concept and then as intervention in

various parts of the manuscript (which is particularly confusing in Q1). Furthermore, most of the sentences in Abstract, Introduction and Methods are in future tense (e.g. L100; L363, L388-389 etc.); please exchange them to past tense. Please see my more detailed comments below. I hope that the Recommender and the Authors will find these comments helpful, and I am looking forward to your answers.

Response 9: Thank you for taking the time to review our Stage 2 manuscript. We appreciate the feedback and we address your comments individually, below.

Comment 10: Abstract does not include methods used, analyses done, results and discussion; as it currently stands it seems to follow the formatting of registered report abstract, at times containing redundant information. Please exchange and adjust accordingly.

Response 10: We apologize for this oversight. We added brief summaries of the specific methods, results and conclusions to the abstract. As well as clarifying up front the analyses that could not be tested so that the reader knows better what to expect. We also changed future tense to past tense throughout.

Comment 11: Throughout the text it would be important to choose labelling to describe a species, e.g. “resilient”, “sensitive”, without mixing it with “flexible”, “successful”, “resistant”, “vulnerable” or “threatened” for clarity.

Response 11: Thank you for pointing this out. We now consistently refer to common species with increasing populations in human-modified areas as disturbance-resilient, and declining species that disappear from human-modified areas as disturbance-sensitive.

Comment 12: Introduction, Research questions, Methods and Supplementary Material part also follows registered report design and include questions that could not be answered due to constraints mentioned above. As this makes it very confusing to the reader, I would suggest to re-write the manuscript focusing on Florida scrub-jays only and questions that could be tested in this species.

Response 12: We understand that it is unusual and we apologize for the confusion. We retained the references to our original study design because part of the Stage 2 process is to faithfully report what was achieved from the peer-reviewed and recommended Stage 1 registered report. However, we will likely take your suggestion after this Stage 2 manuscript is approved when we submit this research to a journal. Since the Stage 2 version will be approved and publicly available at that time, the transparency of the research process can be maintained.

Comment 13: L118-126. This paragraph, while okay for a registered report, is a bit too broad for a Stage 2 report; it should be rewritten to reflect specific steps that this Stage 2 report brings to the research on behavioural flexibility.

Response 13: We edited this paragraph to focus on the specific contributions of the Stage 2 report. It now reads (lines 137-142):

“While our previous research on great-tailed grackles illustrated that flexibility can be increased and can affect behavioral interactions with the environment, the research presented here took the next step by evaluating the generalizability of this technique to a distinct species negatively affected by anthropogenic change. Consequently, the results from this research are important to advance our understanding of the causes and consequences of flexibility by linking behavior to environmental change, cognition, and conservation of declining species.”

Comment 14: Why did the Authors collected data on 14 Florida scrub-jays, as the minimum sample to detect differences as calculated by the power analysis was 15 (L235-239)? This is a bit confusing, and it would be important to provide some justification on how this discrepancy was treated.

Response 14: FLSJ are declining in suburban areas and so it was difficult to find the planned number of individuals (half of the 15 planned minimum sample). We were able to get as close as possible with 6 suburban and 8 natural site individuals, for a total of 14 jays. We now clarify planned and achieved sample sizes for all research questions in a new Table 2 (line 438). Because the minimum sample size was not met for J.Q1, 3 or 4, we add to the first paragraph of the discussion (lines 513-514 and lines 546-551) that our small sample sizes make it difficult to draw firm conclusions from our results.

Comment 15: How many observers collected data on microhabitat and food use, and reversal learning? Did the Authors do some IOR on data collected by different observers or trained the observers to a certain reliability threshold?

Response 15: All data was collected by the first/corresponding author.

Comment 16: As mentioned above, comparisons on reversal learning between 8 birds in the natural areas and only 1 bird in the human modified areas is not balanced; I do not think that any insight can be made from comparisons with only one bird. Can the Authors please justify / explain this in more detail?

Response 16: In J.Q1 we compared reversal learning between 6 suburban and 8 natural site jays (14 total). Because we found no differences in reversal learning between

suburban and natural site jays, we combined foraging and habitat use data across sites for J.Q3 & 4 (now stated in the first paragraph of the Results, lines 448-450). We were unable to conduct the serial reversal learning flexibility manipulation on more than 1 suburban jay, and you are right that it does not make sense to infer anything from a comparison with 1 bird.

Comment 17: J.Q1 Are these analyses on 8 and 1 bird, respectively, or do they include all 14 birds?

Response 17: We apologize for the confusion and now include Table 2 to clarify the sample sizes for all research questions. J.Q1 included 14 jays (6 suburban, 8 natural site jays).

Comment 18: Furthermore, it would be great to clarify whether there are two tasks (one including reversal 0 and 1 and another one including more reversals), as well as what is meant by “baseline associative learning” (L409) and “behavioural flexibility” (L410).

Response 18: Thank you for pointing out this disconnect. We revised the methods section to clarify the 3 phases of the behavioral flexibility methods: Associative learning (Reversal 0), baseline behavioral flexibility (Reversal 1) and serial reversal learning (Reversals 2+) for the flexibility manipulation. We also edited the figure legend accordingly (now in Lines 461-462).

Comment 19: L437-446. First paragraph of the Discussion should summarize the main findings of the current manuscript; now it is perhaps a bit too brief regarding the four questions asked in the paper.

Response 19: Good point. The last portion of this paragraph now also includes a summary of the J.Q1 results and more detail of the J.Q3 & 4 results (lines 508-513).

Comment 20: Minor comments and copy-edits:

Response 20: We incorporated all of your suggested minor edits, below, unless otherwise noted. Thank you for your time reviewing our manuscript! Your input has improved the quality and readability.

L26-28. This sentence does not read clearly, please rephrase.

L30. Can delete “influencing success in” and rephrase “This project” -> “This study”.

L33. “apply them” -> “apply these”. - **We ended up deleting this sentence because it was not specifically relevant to this Stage 2.**

L58-59. A bit unclear, please rephrase.

L110. What do the Authors mean by “focal species”; Florida scrub-jays or other jay species?

L199-201 and Table 1. Section on Q2. Would suggest deleting. - **We are not permitted to delete Q2 for a Stage 2 report, but we will likely take your suggestion for the future publication of this article at a journal.**

L221. The Authors should delete “planned” and talk about the actual sample. - **For the Stage 2, we must retain information on the planned sample size that informed our power analysis and ability to make inferences from our results. We now clarify planned and achieved sample sizes in this section as well as a new Table 2.**

L222. Where are Archbold Biological Station and Lake Placid located? It would be great to have coordinates as well as the size of the location provided.

L226. Why did the Authors collect blood samples from these birds?

L243. Per species or per population?

L246. “but we met or exceeded the minimum sample size for observations before and after the manipulation” -> What was the sample size of birds collected in the end? It would be important to state this explicitly.

L257. Which hours were the observations done in the morning and in the afternoon?

L284. Rephrase to “8am-3pm” for clarity. Furthermore, which months/years did you collect the data (see L395 which mentions two years of data collection)?

L286. What did the Authors consider with “fear behaviours”?

L288. Were 4 sessions both minimum and maximum (with an exception of one bird post-manipulation), or did some birds have more than 4 sessions?

L291-293. This sentence does not read clearly, please rephrase.

L293-296. This is a great approach to make sure there are no potential confounds in refilling of the feeders! - **Thank you so much!**

L297. How far apart were the feeders from each other?

L301-306. Do the Authors mean in 12 trials or less, each or both? So, just to make sure I understand correctly, a reversal was passed when jays had 10/12 correct trials (the total number of trials could have been much higher) and reversals continued until the jays passed two consecutive reversals in 12 or less trials each (i.e. maximum number of trials for two reversals would be 24)? - **Yes, that is correct. We reworded to clarify.**

L333-334. Here it would be great to remind the reader that the analyses are on reversal learning and not behavioural observations.

L365-372. This part is confusing as the question was changed; I would suggest to only include the new analyses as changes to the analyses were stated above. Further, do “disturbance-resilient” and “disturbance-resistant” mean the same thing? If not, please explain the difference. Should L368 Florida scrub-jays read as “disturbance-sensitive” (see also Figure 2)? - **These are both typos! Thank you for catching this!**

L394. “scrub-jay” -> “scrub-jays”

L395. I think “natural” is missing before “areas”.

L397. I guess the Authors mean “reversal learning” and not “research learning”?

L411. “rather than or in this plot” -> I do not understand this part of the sentence. - **Apologies, the mathematical symbols for “phi” and “lambda” did not show up in the pdf.**

L418. “within the disturbance sensitive species, a subset of individuals were resilient to

disturbance” -> do the Authors mean because these individuals were found in human-modified habitats, i.e. suburban population? If so, perhaps just state “within the disturbance-sensitive species, a subset of individuals lived in human-modified environments”.

L442-444. Description of grackle may be better merged with the previous sentence (L440-442).

L447. “coincident” -> “coincidence”

L449. “significantly” -> “substantially” (significant is usually used in the context of statistical significance).

With best wishes,
Vedrana Šlipogor

Thank you for taking the time to give feedback on our Stage 2, Vedrana!

Review by Alizée Vernouillet, 17 Apr 2026 00:00

Comment 21: The current study focuses on the influence of a behavioral flexibility manipulation (through a spatial reversal learning approach involving two feeders) on the foraging and micro-habitats diversity in Florida scrub jays living in natural areas and in suburban areas. This project is part of a larger research program (ManyIndividuals) that was submitted as a registered program in PCI and evaluated in 2022.

As mentioned before, the project is very intriguing and with a lot of potential. I understand that this project was ambitious and clearly the authors have put a lot of efforts into the project. However, due to limited time and resources, the current paper presents a subset of the original planned ideas and research questions. Mainly, the study was done on one vulnerable species, the Florida scrub jay (and not on multiple species, including the California scrub jay and the blue jay), so this meant that the main research question Q2 was not addressed. The number of individuals tested was also lower than planned (14 instead of 15 for Q1, 9 instead of 20 for Q3 and Q4). Because of these limitations, the current versions of the abstract, introduction, and research questions sections taken directly from Stage 1 are slightly misleading. The fact that only Florida scrub jays were tested need to be made clearer, both in the introduction and in the research questions, as a lot of emphasis was put on the species comparison, especially on the importance of comparing species that adapt well to the urban environments and those that do not.

Response 21: Thank you for taking time to review the Stage 2 manuscript! We appreciate your insights throughout the registered report process.

We apologize that these sections were misleading. We now more explicitly detail throughout that we were not able to study all planned species or obtain the planned sample sizes. Moreover, we created a new Table 2 (lines 438-441) that lists the 4 research questions, the planned species and sample sizes as well as the achieved species and sample sizes.

Comment 22: Related to the previous point, the abstract needs to be updated to reflect these limitations and include results and conclusions of the study. Likewise, the parts taken from Stage 1 (abstract, introduction, research questions, supplementary material) needs to be changed to past tense.

Response 22: We updated the Abstract to be past tense, including specific methods, results and conclusion for this study. We tried to find and edit all parts that were present or future tense into past tense.

Comment 23: The new analysis done to address Q2 (i.e., the comparison between the suburban and the rural populations) needs a stronger rationale. Given that there are more micro-habitats described in the human-modified areas than in natural areas, jays in human-modified areas have more options on where to go and would be expected to use more micro-habitats than those in the natural areas due to differences in availabilities. Rather, 1) whether jays use all micro-habitats available in the area they live in (and so, report the descriptive stats) and 2) if their use of each micro-habitat differ from their availability in the area they were observed would provide a stronger approach. For instance, would the birds in the suburban populations use mostly the human-modified micro-habitats or would still favour the natural ones? Similarly, using proportions rather than simply number of different food items could provide a more interesting comparison between the two populations tested.

Response 23: We revised the organization of the *unregistered analysis* so that it is distinct from J.Q2, with its own rationale, and more explicitly noted as exploratory. We recognize the validity of your suggestion to control for the availability of microhabitat type when comparing use. However, this was a post-hoc exploratory analysis and so we do not have data on total availability of each microhabitat type in suburban and natural areas. What these data show is that scrub-jays can use human modified microhabitats when they are available, despite having the option to exclusively use remnant native scrub patches. We include a new figure (lines 492-495) showing the proportion of microhabitats and food categories used by jays at each site type, and we emphasized in the discussion that future research could more explicitly quantify a *preference* for human modified microhabitats (lines 531-539).

Comment 24: Additionally, the limitations due to lower sample size need to be acknowledged more explicitly in the discussion, especially when assessing the influence of the manipulation on measures of behavioral flexibility in the wild between the suburban and the rural populations (e.g., l. 496).

Response 24: We now state at the end of the first discussion paragraph that our small sample sizes make it difficult to draw firm conclusions from our results (lines 513-514). Moreover, in the paragraph with the line number you indicate, we added several

sentences acknowledging the small sample size and the consequences of that for our inferences about the manipulation effect (lines 546-551).

Comment 25: Finally, the discussion could include additional elements, such as suggestions as to how to assess whether the flexibility training was too easy (e.g., by increasing the number of choices during the reversal learning task),

Response 25: Good point, we added more to the discussion about increasing the number of feeders (a spatial array rather than just two) to increase the difficulty of the task (lines 570-572).

I have some minor additional comments that can be found below:

Comment 26: Abstract

I. 28 – please change “it primarily trains a specific behavior to address one potential threat TO improve individual success in the wild.”

I. 33 – it is currently unclear what “This” refers to

I. 34 – please change into “of our understanding” or “of our current knowledge”

I. 35 – at this point in the manuscript, it is unclear what are considered flexible species. For instance, was it previously tested or was it simply assumed since they were successful at establishing themselves in cities?

I. 39 – please remove “will” and generally throughout the manuscript, remove instances of future tense

Response 26: We removed some of these sentences from the Abstract in the process of edits for other reviewers. However, we believe all edits also serve to address your comments.

Comment 27: Registered Report Details

I. 52 – at this point in the manuscript, it is unclear what a “disturbance-sensitive” species is.

I. 53 – “habitatS”

Response 27: We made edits to address both of these comments.

Introduction

Comment 28: l. 98 – because flexibility is used to refer to multiple concepts in the intro, it is hard to know what it means here. I think you mean reversal learning improved species performance or speed to change to a different response, but not general flexibility as this is one of the main aspects you want to test this later. This specificity is important given that the entire rationale of being one of the firsts to test whether a manipulation can improve a species' behavioral flexibility as a general cognitive ability.

Response 28: We rephrased this to “the time it took to change their behavior” (line 113)

Comment 29: l. 101 – at this point in the manuscript, saying “California scrub-jays OR blue jays” feels awkward as it gives the impression that the study had not been done yet. Please see main comment

Response 29: We added to this paragraph (lines 115-122) to make it more clear that while we aimed to test these other, disturbance-resilient species, we ultimately were not able to within the timeframe of the funding. However, because we know the flexibility manipulation affects behavior of disturbance-resilient grackles, our study still includes important information about the generalizability of this method to declining (disturbance-sensitive) species.

Comment 30: l. 102 – here, Florida scrub jays are presented as vulnerable, but later in the manuscript it is reported as threatened. Please make it consistent throughout.

Response 30: Apologies, this is a mix of ESA and IUCN listing. We changed to describe it as “threatened” throughout (ESA listing).

Comment 31: l. 103 – change “do” into “did”

l. 110 – you can make it clearer by adding FSCJ

l. 111 – this sentence is a bit unclear. How about changing “In the event that social learning is not used by a given population to spread the behaviors of manipulated individuals” into “even if the behaviors of manipulated individuals does not spread in the population through social learning”?

Response 31: We made these changes.

Research questions

Comment 32: l. 139-140 – the link between threatened species and the success to human modified environments needs to be clearer in the introduction or explained more in details here.

Currently, the introduction focuses on the idea that species that are successful at adapting to a human modified environment might be more flexible. However, there is no rationale nor reported evidence from previous studies that species that are less successful at adapting to new environments are less flexible. Hence, this statement sounds overreaching from the predictions that are being made.

Response 32: Modifying the main research questions from a programmatic registered report after the results are known is discouraged. However, we added to the introduction the following sentence in lines 91-93 to highlight that species that are less successful at adapting to new environments are less flexible:

“Globally, bird species with more diverse foraging behavior in the wild were less likely to be threatened with extinction (Ducatez et al., 2020), supporting the idea that the ability to flexibly adapt to new foraging challenges reduces population decline.”

Comment 33: l. 148-150 – this sentence is quite redundant “if flexibility is not manipulatable (...), this would indicate that (...) flexibility is not trainable”

l. 152 – the term “association” does not clearly indicate that it refers to when the species originally adapted to human modified environments. Maybe it can be replaced into “when originally adapting to human modified environments”?

Response 33: We apologize for the lack of clarity in these lines. Unfortunately, changing these main research questions from this programmatic registered report is discouraged.

Comment 34: Figure 2. Please credit where the data for the three species distribution comes from.

l. 170 – it might be worth adding that FLSJ are not found in urban areas to make the contrast with the two other species more evident (or by moving the sentences at l.177-178 up). If the rationale about these two other species is being kept, it might be useful to add that only FLSJ were tested here.

l. 188-190 – as it was previously mentioned (l. 103-105) that the potential spread of the effects of the manipulation would not be investigated, this statement feels out of place.

l. 195 – here, it might be worth to remind the reader that it is evaluated within a species

Response 34: We made these changes.

Methods

Comment 35: l. 227 – was the success measures assessed after having the individuals were banded?

Response 35: Yes, the behavioral observations to quantify success measures occurred after individuals were banded. Some individuals were banded years before this study started. However, at minimum, behavioral observations occurred one week after banding, but researchers wore different clothes and a face mask during banding to prevent recognition.

Comment 36: How was potential displacement or disturbance (by another jay or by another species including humans) was accounted when doing observations? Would the observation stop?

Response 36: All social interactions (aggressive, affiliative, conspecific, heterospecific) were noted during observations. These are a natural part of the jays' lives that trade off with foraging and habitat use. They are therefore important to capture during behavioral observations and we did not stop observations when this occurred.

Comment 37: l. 240 – what was the sampling interval?

Response 37: We waited at least 24 hours between behavioral observation sessions. This is now added to lines 263-264.

Comment 38: l. 284 – please use international time system

Response 38: We now use the 24 hour time system

Comment 39: l. 287 – were the baseline observations of microhabitats use and food items consumed done prior to the baseline/feeder habituation?

Response 39: Yes, this is now clarified in lines 299-300.

Comment 40: l. 300 – were the feeders removed outside of testing and reversal sessions?

Response 40: Yes, this is now clarified in lines 303-304.

Comment 41: l. 373 and l. 383 – please spell out the entire research question

Response 41: Done.

Comment 42: Results

l. 397 – word(s) missing before “areas”

l. 411 – word(s) missing after “rather than”

Response 42: We addressed these issues.

Discussion

Comment 43: l. 452 – the term “congener” feels strange and could be replaced by “closely-related species”

l. 460 – this would be an overclaim (see comment earlier)

Response 43: We made these changes

Comment 44: l. 469 – mentioning the ceiling value of the different measures of success in the methods/results of would strengthen this argument. As it is written, it is unclear whether the lack of significant results could simply be due to the lack of “improvement” possible (as developed in the following paragraph).

Response 44: We added to the discussion in lines 579-586 to describe why it is unlikely that the lack of behavioral change after the manipulation is due to a ceiling effect of number of possible food types and microhabitats.

Comment 45: l. 470 – it would be clearer to state that the birds improved behavioral flexibility in the context of the task, not generally

Response 45: We made this change

Comment 46: l. 472 – when reporting the results with the grackles, it is important to highlight that the manipulation did improve foraging-related behaviors but not micro-habitat use. This could suggest that the manipulation might have influenced some of the FLJA behaviors, but not the ones assessed (or at least, now how they were measured/analysed).

Response 46: Good point! We made edits to this effect in the first paragraph of the discussion in lines 504 and in lines 137-140.

Comment 47: I hope my comments are relevant and can help improve the manuscript! I am really happy to see the results of this project coming together and congratulate you once again for this very interesting study.

Response 47: Thank you so much for the time and consideration you gave to our Stage 1 and Stage 2 manuscripts!

Dr. Alizée Vernouillet